POWERING POTENTIAL

Data Analytics Lifecycle

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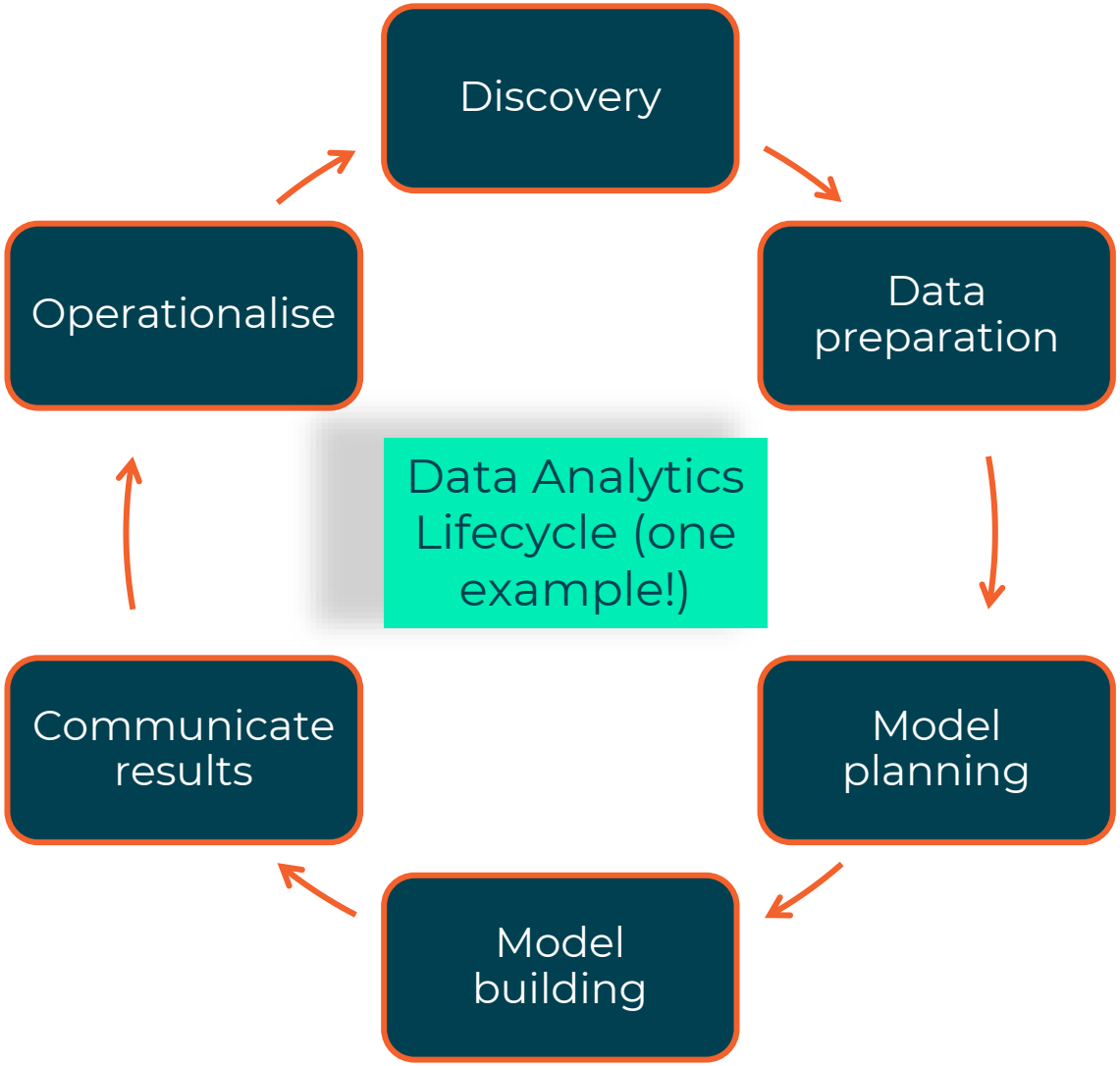
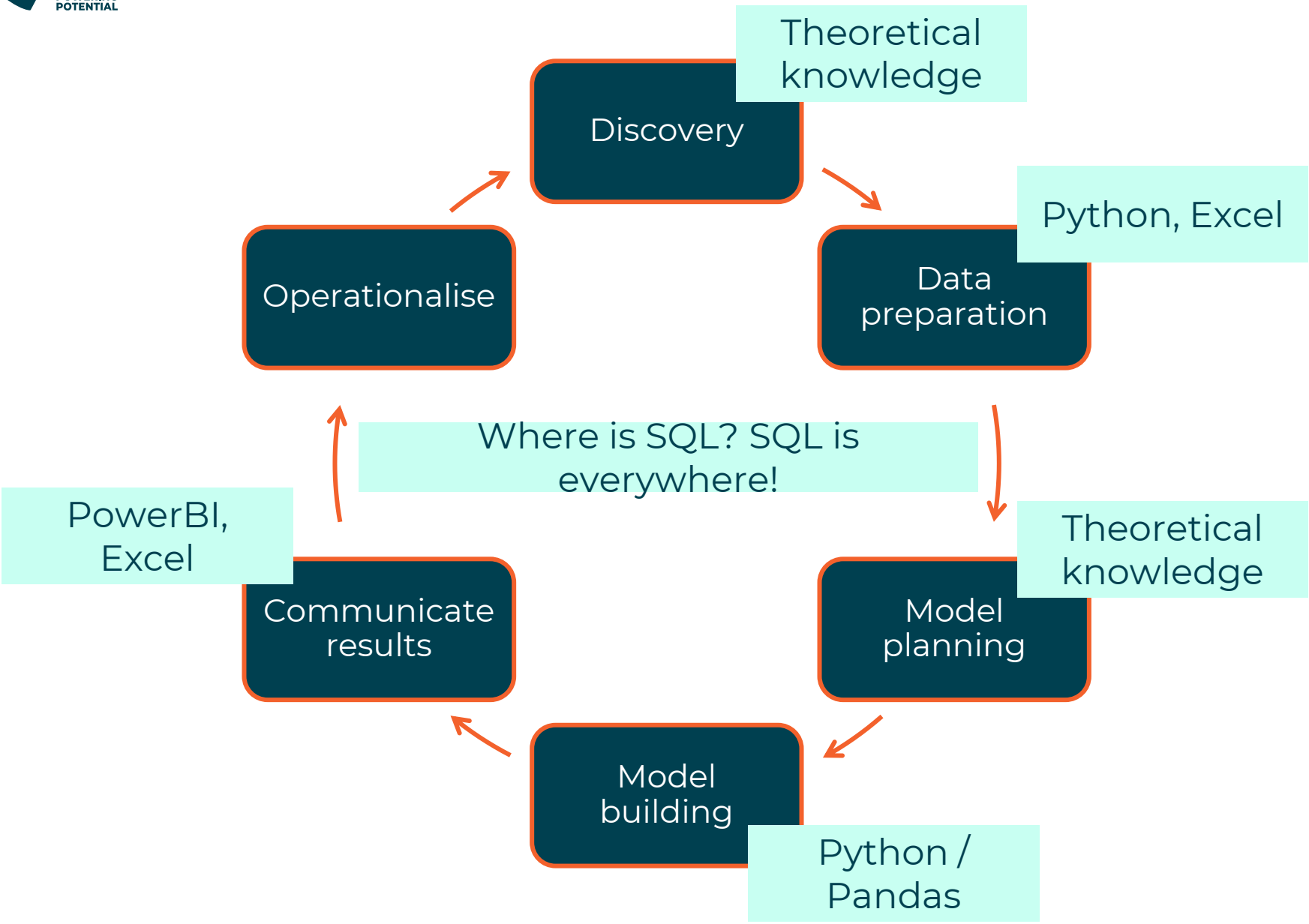
LESSON AGENDA

1	Discovery
2	Data preparation
3	Model planning
4	Model building
5	Communicate results
6	Operationalise

Data Analytics Lifecycle



Data Analytics Lifecycle



POWERING POTENTIAL

Phase 1 – Discovery

Phase 1—Discovery

→Are there any previous similar projects?

→What are the available resources?

people, technology, time and data

→Become familiar with the data - test the data

→What is the initial hypothesis (IH) ?

Understand the objectives!

Phase 2 – Data preparation

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→ETL or ELT or ETLT
→Requires analytic sandbox
→Perform analytics
→Construct final dataset from raw data
→**Data cleansing**

Prepare the data!

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What's a sandbox?

- "Data scientists playground"
- BI tool used for discovery and situational analysis
- Data access and delivery – gathering and integration of data from a variety of data sources and types (internal/external data, structured/unstructured data)

Prepare the data!

Phase 3– Model planning

Phase 3 – Model planning

→ Determine the methods, techniques, and workflow intended for use

→ Explore the data to identify relationships between variables

→ Select most suitable models

Select suitable models!

Phase 4 – Model building

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→Develop datasets for testing, training, and production purposes

→Build and execute models based on the work done in the model planning

→Qualify environment for model/workflow execution

Build the model!

Phase 5 – Communicate results

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→Identify key findings
→Quantify the business value
→Establish missing or insufficiently covered issues
→Report to summarise and convey findings to stakeholders

Summarise and communicate!

Phase 6 – Operationalise

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→Deliver final reports, briefings, code, and technical documents

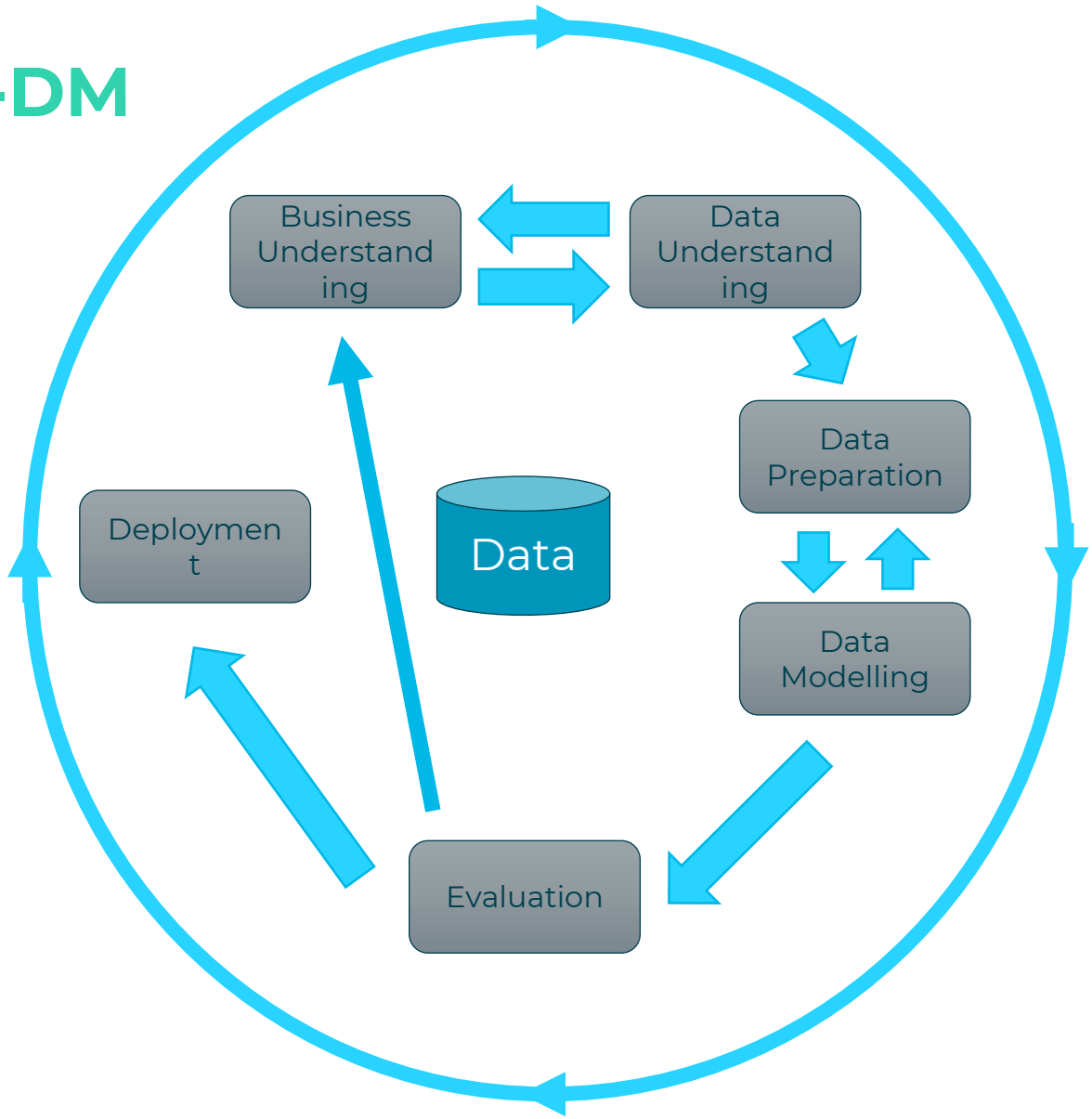
→Run a pilot project to implement the models in a production environment.

Final delivery!

Alternative lifecycle models

Data Analytics Lifecycle

CRISP-DM



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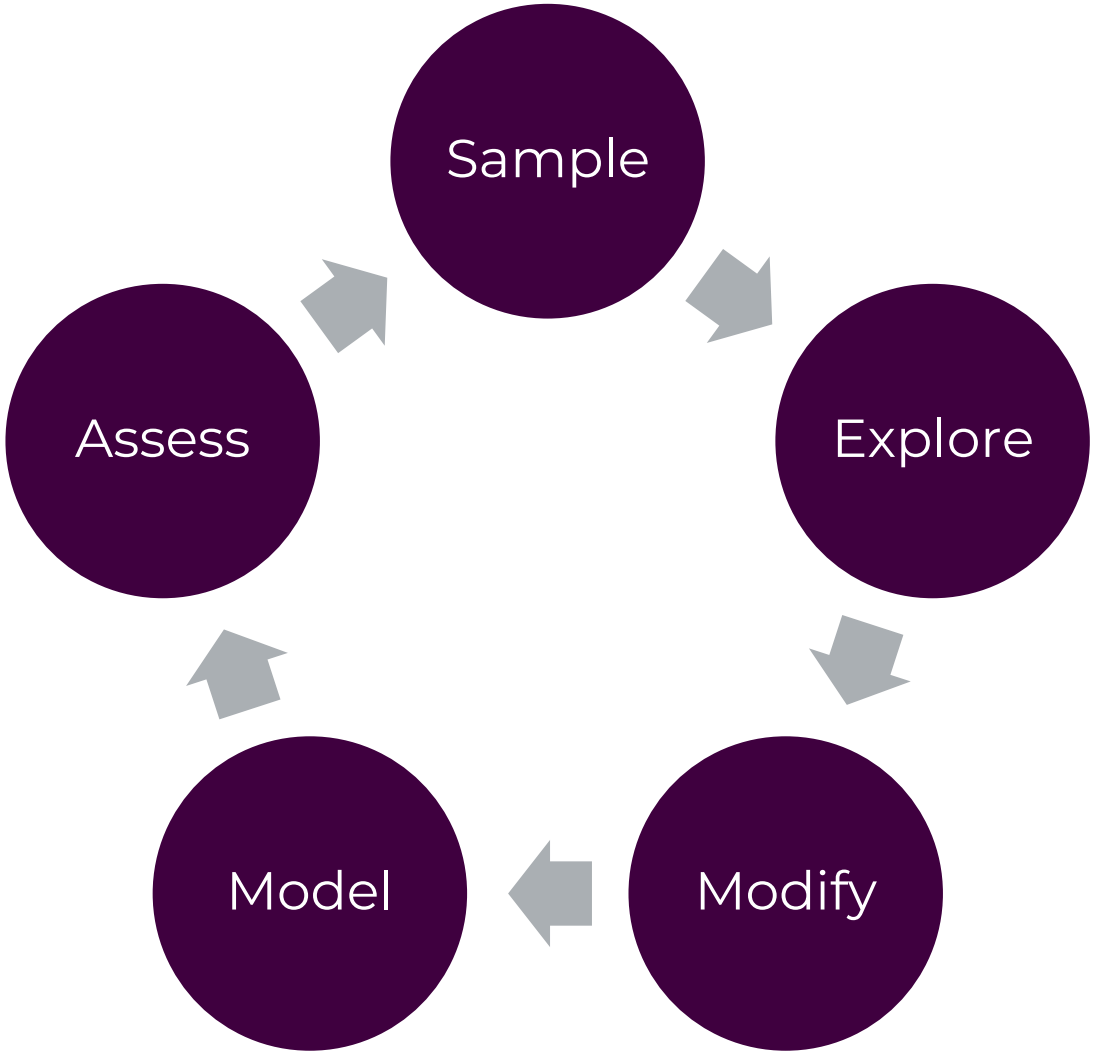
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